50 SQL Queries




```
SELECT  *
FROM  employees
WHERE  salary > 50000;
```



01. How to find the second-highest salary?

```
SELECT MAX(salary) AS second_highest_salary
FROM employees
WHERE salary < (SELECT MAX(salary) FROM employees);
```

02. How to find the highest salary in each department?

```
SELECT department_id, MAX(salary) AS highest_salary
FROM employees
GROUP BY department_id;
```

03. How to count the total number of employees?

```
SELECT COUNT(*) AS total_employees
FROM employees;
```

04. How to find employees earning more than ₹50,000?

```
SELECT employee_id, name, salary
FROM employees
WHERE salary > 50000;
```

05. How to find the average salary of all employees?

```
SELECT AVG(salary) AS average_salary
FROM employees;
```

06. How to find employees whose names start with 'A'?

```
SELECT *
FROM employees
WHERE name LIKE 'A%';
```

07. How to find employees whose names end with 'n'?

```
SELECT *
FROM employees
WHERE name LIKE '%n';
```

08. How to find employees who joined in 2025?

```
SELECT *
FROM employees
WHERE YEAR(join_date) = 2025;
```

09. How to find the lowest salary in the company?

```
SELECT MIN(salary) AS lowest_salary
FROM employees;
```

10. How to find employees who have no manager?

```
SELECT *
FROM employees
WHERE manager_id IS NULL;
```

11. How to find the third-highest salary?

```
SELECT DISTINCT salary
FROM employees
ORDER BY salary DESC
LIMIT 1 OFFSET 2;
```

12. How to find duplicate employee names?

```
SELECT name, COUNT(*) AS total
FROM employees
GROUP BY name
HAVING COUNT(*) > 1;
```

13. How to find employees earning above the average salary?

```
SELECT *
FROM employees
WHERE salary > (
    SELECT AVG(salary)
    FROM employees
);
```

14. How to find the number of employees in each department?

```
SELECT department_id, COUNT(*) AS total_employees
FROM employees
GROUP BY department_id;
```

15. How to find the employee with the highest salary?

```
SELECT *
FROM employees
WHERE salary = (
 SELECT MAX(salary)
 FROM employees
);
```

16. How to find employees who joined in the same year as their manager?

```
SELECT e.employee_id, e.name
FROM employees e
JOIN employees m
ON e.manager_id = m.employee_id
WHERE YEAR(e.join_date) = YEAR(m.join_date);
```

17. How to find the department with the highest average salary?

```
SELECT department_id, AVG(salary) AS avg_salary
FROM employees
GROUP BY department_id
ORDER BY avg_salary DESC
LIMIT 1;
```

18. How to find employees who earn more than their manager?

```
SELECT e.name, e.salary, m.name AS manager_name, m.salary
AS manager_salary
FROM employees e
JOIN employees m
ON e.manager_id = m.employee_id
WHERE e.salary > m.salary;
```

19. How to find the second-highest salary in each department?

```
SELECT department_id, MAX(salary) AS second_highest
FROM employees e
WHERE salary < (
    SELECT MAX(salary)
    FROM employees
    WHERE department_id = e.department_id
)
GROUP BY department_id;
```

20. How to find departments having more than 5 employees?

```
SELECT department_id, COUNT(*) AS total_employees
FROM employees
GROUP BY department_id
HAVING COUNT(*) > 5;
```

21. How to find employees who joined in the last 30 days?

```
SELECT *
FROM employees
WHERE join_date >= CURRENT_DATE - INTERVAL 30 DAY;
```

22. How to find the highest-paid employee in each department?

```
SELECT *
FROM employees e
WHERE salary = (
    SELECT MAX(salary)
    FROM employees
    WHERE department_id = e.department_id
);
```

23. How to count employees whose names start and end with the same letter?

```
SELECT COUNT(*) AS total_employees
FROM employees
WHERE LEFT(name, 1) = RIGHT(name, 1);
```

24. How to find employees who don't belong to any department?

```
SELECT *
FROM employees
WHERE department_id IS NULL;
```

25. How to find the total salary paid by each department?

```
SELECT department_id, SUM(salary) AS total_salary
FROM employees
GROUP BY department_id;
```

26. How to find employees with the same salary?

```
SELECT salary, COUNT(*) AS total_employees
FROM employees
GROUP BY salary
HAVING COUNT(*) > 1;
```

27. How to find the top 3 highest-paid employees?

```
SELECT *
FROM employees
ORDER BY salary DESC
LIMIT 3;
```

28. How to find the department with the most employees?

```
SELECT department_id, COUNT(*) AS total_employees
FROM employees
GROUP BY department_id
ORDER BY total_employees DESC
LIMIT 1;
```

29. How to find employees who joined before their manager?

```
SELECT e.name AS employee,
       e.join_date AS employee_join_date,
       m.name AS manager,
       m.join_date AS manager_join_date
FROM employees e
JOIN employees m
ON e.manager_id = m.employee_id
WHERE e.join_date < m.join_date;
```

30. How to find the Nth-highest salary?

```
SELECT DISTINCT salary
FROM employees
ORDER BY salary DESC
LIMIT 1 OFFSET N-1;
```

31. How to find employees who earn the same salary as their manager?

```
SELECT e.name AS employee,
       m.name AS manager,
       e.salary
FROM employees e
JOIN employees m
ON e.manager_id = m.employee_id
WHERE e.salary = m.salary;
```

32. How to find employees who joined before 2020?

```
SELECT employee_id, name, join_date
FROM employees
WHERE join_date < '2020-01-01';
```

33. How to find the total salary of employees in each department?

```
SELECT department_id,
       SUM(salary) AS total_salary
FROM employees
GROUP BY department_id;
```

34. How to find departments where the average salary is greater than 60,000?

```
SELECT department_id,
       AVG(salary) AS avg_salary
FROM employees
GROUP BY department_id
HAVING AVG(salary) > 60000;
```

35. How to find employees whose salary is between 40,000 and 80,000?

```
SELECT employee_id, name, salary
FROM employees
WHERE salary BETWEEN 40000 AND 80000;
```

36. How to find employees who belong to the IT department?

```
SELECT e.employee_id, e.name
FROM employees e
JOIN departments d
ON e.department_id = d.department_id
WHERE d.department_name = 'IT';
```

37. How to find employees who don't have a manager?

```
SELECT employee_id, name
FROM employees
WHERE manager_id IS NULL;
```

38. How to find the number of employees hired in each year?

```
SELECT YEAR(join_date) AS joining_year,
       COUNT(*) AS total_employees
FROM employees
GROUP BY YEAR(join_date)
ORDER BY joining_year;
```

39. How to find the maximum salary in each department?

```
SELECT department_id,
       MAX(salary) AS highest_salary
FROM employees
GROUP BY department_id;
```

40. How to find employees whose salary is higher than everyone in department 10?

```
SELECT employee_id, name, salary
FROM employees
WHERE salary > ALL (
    SELECT salary
    FROM employees
    WHERE department_id = 10
);
```

41. How to find employees who have the same manager?

```
SELECT manager_id,
       COUNT(*) AS total_employees
FROM employees
WHERE manager_id IS NOT NULL
GROUP BY manager_id
HAVING COUNT(*) > 1;
```

42. How to find employees with the longest names?

```
SELECT employee_id, name
FROM employees
WHERE LENGTH(name) = (
    SELECT MAX(LENGTH(name))
    FROM employees
);
```

43. How to find employees whose names contain 'an'?

```
SELECT employee_id, name
FROM employees
WHERE name LIKE '%an%';
```

44. How to find the total number of employees in each job role?

```
SELECT job_title,
       COUNT(*) AS total_employees
FROM employees
GROUP BY job_title;
```

45. How to find the department with the lowest average salary?

```
SELECT department_id,
       AVG(salary) AS avg_salary
FROM employees
GROUP BY department_id
ORDER BY avg_salary ASC
LIMIT 1;
```

46. How to rank employees based on their salary?

```
SELECT employee_id,
       name,
       salary,
       RANK() OVER (ORDER BY salary DESC) AS salary_rank
FROM employees;
```

47. How to find the top 3 salaries in each department?

```
SELECT *
FROM (
    SELECT employee_id,
           name,
           department_id,
           salary,
           DENSE_RANK() OVER (
               PARTITION BY department_id
               ORDER BY salary DESC
           ) AS salary_rank
    FROM employees
) e
WHERE salary_rank <= 3;
```

48. How to find each employee's salary difference from the department average?

```
SELECT employee_id,
       name,
       department_id,
       salary,
       salary - AVG(salary) OVER (
           PARTITION BY department_id
       ) AS salary_difference
FROM employees;
```

49. How to find the highest-paid employee in the company using a window function?

```
SELECT employee_id, name, salary
FROM (
    SELECT employee_id,
           name,
           salary,
           RANK() OVER (ORDER BY salary DESC) AS rnk
    FROM employees
) e
WHERE rnk = 1;
```

50. How to find the second-highest salary using DENSE_RANK?

```
SELECT employee_id, name, salary
FROM (
    SELECT employee_id,
           name,
           salary,
           DENSE_RANK() OVER (
               ORDER BY salary DESC
           ) AS salary_rank
    FROM employees
) e
WHERE salary_rank = 2;
```

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